PIZZA SALES ANALYSIS REPORT

1. Project Overview: -

The Pizza Sales Analysis project focuses on analyzing pizza sales data using SQL and Power BI. It helps understand overall sales performance, revenue, customer orders, and popular pizza categories. The project includes interactive dashboards to identify sales trends, best-selling and worst-selling pizzas, and customer ordering patterns. These insights help businesses make better decisions and improve sales performance.

2. Dataset Overview: -

The Pizza Sales dataset contains transaction-level sales information for the period January 1, 2015, to December 31, 2015. It provides details about customer orders, pizza products, categories, sizes, quantities and pricing. The dataset is used to analyze sales performance, revenue distribution, ordering patterns and product popularity.

Column Name	Description
pizza_id	Unique pizza record ID
order_id	Unique order ID
order_date	Date of the order
order_time	Time of the order
pizza_name	Name of the pizza
pizza_category	Category of pizza
pizza_size	Size of pizza
quantity	Number of pizzas ordered
unit_price	Price per pizza
total_price	Total sales amount

Dataset Highlights:

- Time period: January–December 2015.
- Contains order-level and product-level sales information.
- Includes four pizza categories: Classic, Supreme, Chicken and Veggie.
- Includes multiple pizza sizes.
- Supports revenue, quantity, order and product performance analysis.

3. Data Analysis Using SQL Server (Business Transactions): -

A.KPI's: -

We performed structured analysis in PostgreSQL to answer key business questions:

1. TOTAL REVENUE: - Evaluate the overall sales value generated through pizza orders.

Results Messages

	Total_Revenue
1	817860.05083847

2. AVERAGE ORDER VALUE (AOV): - Determine the average sales revenue earned from each customer order.

Results Messages

	Avg_order_Value
1	38.3072623343546

3. TOTAL PIZZAS SOLD: - Determine the total number of pizzas sold across all customer orders.

Results Messages

	Total_pizza_sold
1	49574

4. TOTAL ORDERS: - Determine the total number of unique orders placed by customers.

Results Messages

	Total_Orders
1	21350

5. AVERAGE PIZZAS PER ORDER: - Determine the average number of pizzas ordered per customer order.

Results Messages

	Avg_Pizzas_per_order
1	2.32

B. DAILY TREND FOR TOTAL ORDERS: -

Analyze the total number of orders received on each day of the week to identify the busiest days.

Results Messages

	order_day	total_orders
1	Saturday	3158
2	Wednesday	3024
3	Monday	2794
4	Sunday	2624
5	Friday	3538
6	Thursday	3239
7	Tuesday	2973

C. MONTHLY TREND FOR TOTAL ORDERS: -

Analyze the total number of orders received each month to identify monthly sales patterns.

	Month_Name	Total_Orders
1	February	1685
2	June	1773
3	August	1841
4	April	1799
5	May	1853
6	December	1680
7	January	1845
8	September	1661
9	October	1646
10	July	1935
11	November	1792
12	March	1840

D. SALES PERFORMANCE BY PIZZA CATEGORY: -

Analyze the total revenue and percentage contribution of each pizza category to overall sales.

Results Messages

	pizza_category	total_revenue	PCT
1	Classic	220053.10	26.91
2	Chicken	195919.50	23.96
3	Veggie	193690.45	23.68
4	Supreme	208197.00	25.46

E. SALES PERCENTAGE BY PIZZA SIZE: -

Analyze the total revenue and percentage contribution of each pizza size to overall sales.

Results Messages

	pizza_size	total_revenue	PCT
1	L	375318.70	45.89
2	M	249382.25	30.49
3	S	178076.50	21.77
4	XL	14076.00	1.72
5	XXL	1006.60	0.12

F. TOTAL PIZZAS SOLD BY CATEGORY IN FEBRUARY: -

Analyze the total number of pizzas sold in each category during February to identify category-wise performance.

Results Messages

	pizza_category	Total_Quantity_Sold
1	Classic	14888
2	Supreme	11987
3	Veggie	11649
4	Chicken	11050

G. TOP 5 PIZZAS BY REVENUE: -

Identify the top five pizzas that generate the highest revenue for the business.

Results Messages

	pizza_name	Total_Revenue
1	The Thai Chicken Pizza	43434.25
2	The Barbecue Chicken Pizza	42768
3	The California Chicken Pizza	41409.5
4	The Classic Deluxe Pizza	38180.5
5	The Spicy Italian Pizza	34831.25

H. BOTTOM 5 PIZZAS BY REVENUE: -

Identify the five pizzas that generate the lowest revenue for the business.

	pizza_name	Total_Revenue
1	The Brie Carre Pizza	11588.4998130798
2	The Green Garden Pizza	13955.75
3	The Spinach Supreme Pizza	15277.75
4	The Mediterranean Pizza	15360.5
5	The Spinach Pesto Pizza	15596

I. TOP 5 PIZZAS BY QUANTITY SOLD: -

Identify the five most popular pizzas based on the total quantity sold to customers.

	pizza_name	Total_Pizza_Sold
1	The Classic Deluxe Pizza	2453
2	The Barbecue Chicken Pizza	2432
3	The Hawaiian Pizza	2422
4	The Pepperoni Pizza	2418
5	The Thai Chicken Pizza	2371

J. BOTTOM 5 PIZZAS BY QUANTITY SOLD: -

Identify the five least popular pizzas based on the total quantity sold to customers.

Results Messages

	pizza_name	Total_Pizza_Sold
1	The Brie Carre Pizza	490
2	The Mediterranean Pizza	934
3	The Calabrese Pizza	937
4	The Spinach Supreme Pizza	950
5	The Soppressata Pizza	961

15. TOP 5 PIZZAS BY TOTAL ORDERS: -

Identify the five pizzas that are included in the highest number of unique customer orders.

Results Messages

	pizza_name	Total_Orders
1	The Classic Deluxe Pizza	2329
2	The Hawaiian Pizza	2280
3	The Pepperoni Pizza	2278
4	The Barbecue Chicken Pizza	2273
5	The Thai Chicken Pizza	2225

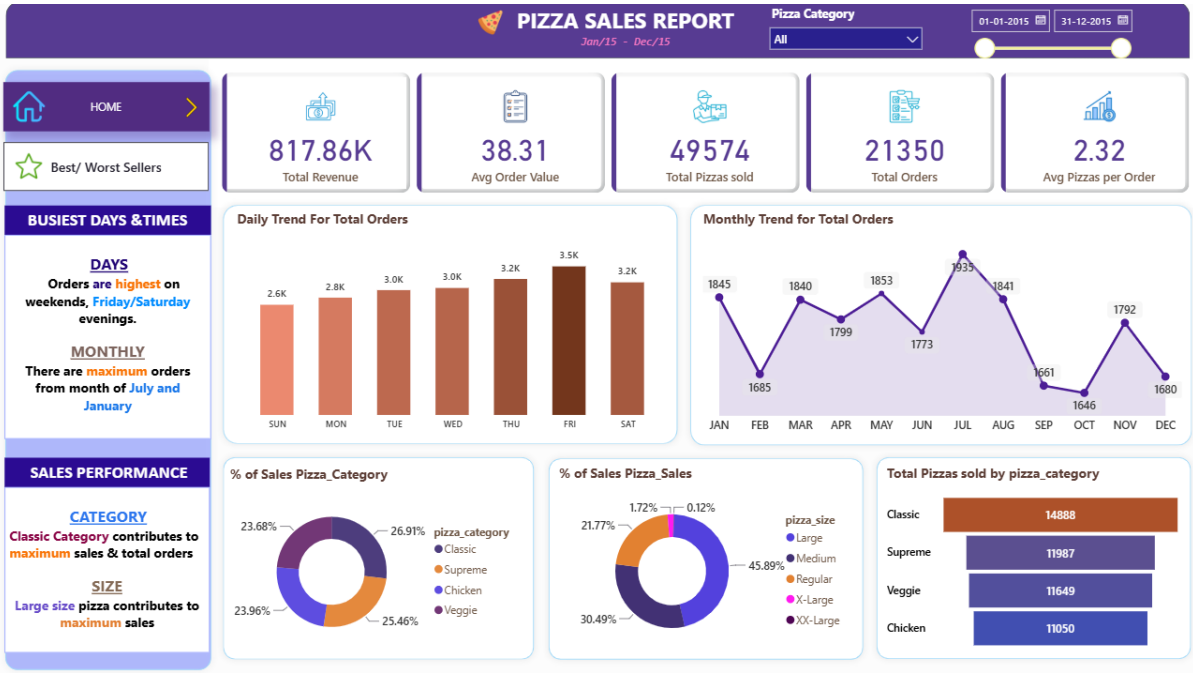
16. BOTTOM 5 PIZZAS BY TOTAL ORDERS: -

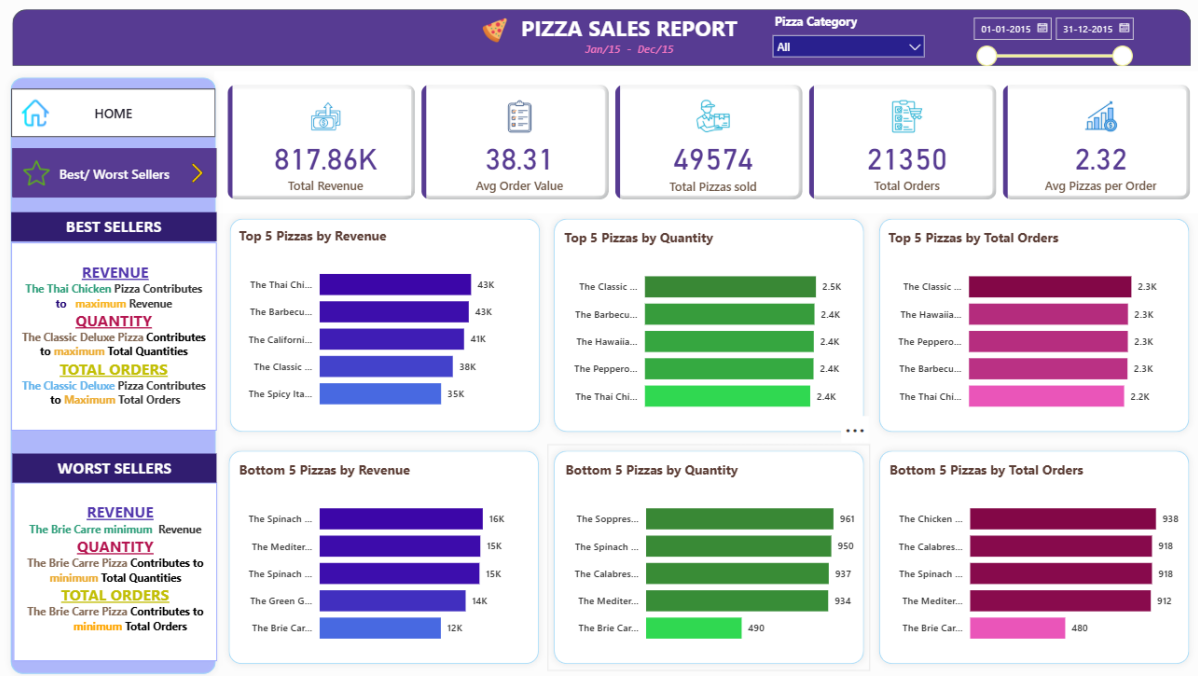
Identify the five pizzas that are included in the lowest number of unique customer orders.

	pizza_name	Total_Orders
1	The Brie Carre Pizza	480
2	The Mediterranean Pizza	912
3	The Spinach Supreme Pizza	918
4	The Calabrese Pizza	918
5	The Chicken Pesto Pizza	938

3. Dashboard in PowerBI: -

Finally, we built an interactive dashboard in Power BI to present insights visually.





4.Business Recommendations: -

Key Insights:

1. Total revenue generated: 817.86K from 21,350 orders.
2. Friday is the busiest day, with approximately 3.5K orders.
3. July records the highest monthly orders (1,935).
4. Classic pizza contributes the highest category sales (26.91%).
5. Large pizzas are the most popular size (45.89%).
6. Thai Chicken generates the highest revenue (43K).
7. Classic Deluxe leads in pizza quantity and total orders.
8. Brie Carre is the worst-performing pizza, generating only 12K revenue.

Business Recommendations:

1. Promote popular pizzas such as Thai Chicken and Barbecue Chicken.
2. Introduce Friday special offers to increase sales.
3. Focus on Large and Medium pizza promotions.
4. Create combo deals featuring Classic Deluxe.
5. Improve marketing for low-performing Veggie pizzas.
6. Analyze customer feedback to understand why Brie Carre has low demand.
7. Introduce special discounts during low-demand months, especially October.
8. Focus on increasing average order value through family combos and add-on offers.

Conclusion: -

The analysis shows that Classic pizzas, Large sizes, and Friday orders contribute significantly to business performance. The company should focus on promoting best-selling pizzas, improving low-performing products, and introducing targeted offers to increase sales and revenue.